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Sistematical evaluation of digital wavelet filter selection for identification system, case study: DC arcing detection system

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ABSTRACT

The Discrete Wavelet Transform (DWT) is a valuable tool for analyzing waves with nonstationary disturbances, yet selecting the most appropriate wavelet type remains a significant challenge. This research presents a systematic evaluation approach to identify the optimal wavelet type, using Series Arc Faults (SAFs) as a case study. The study evaluates several DWT filter techniques, including Haar Level 4, Daubechies (Db) Level 4, Symlet Level 4, and Coiflet Level 4, establishing uniform SAF threshold values for each wavelet type. The analysis of SAF data was conducted using the center of gravity (CoG) method and feature extraction, with results illustrated in a radar diagram. The findings revealed that the Db 4 method exhibited the highest accuracy in detecting SAFs, surpassing the other wavelet types by 4%. This indicates that the CoG and feature extraction methods are effective for determining the optimal wavelet type in wave analysis.

1. Introduction

Photovoltaic (PV) systems are increasingly recognized as a rapidly expanding source of renewable energy, driven by the growing demand for sustainable electricity. Indonesia aims to achieve a solar power capacity of 5GW by the year 2020 [1]. The International Energy Agency (IEA) predicts significant growth in PV development [2], with installation costs expected to decrease from $1.1 per watt in 2023 to $0.9 per watt in 2024. Nevertheless, PV installations face potential disruptions, such as arcing (sparks), which may lead to fires or diminish system efficiency [3]. Protection against Arc failure is highly recommended for system reliability [4].

According to [5], the failure of photovoltaic (PV) systems is linked to undetected electrical arcing. Timely detection of arcing is crucial for ensuring system reliability and safety. Identifying low voltage arcing within these systems is challenging due to the resemblance of the signal characteristics to noise and normal power variations [6]. Arcing can manifest at low voltages, particularly at a minimum PV voltage of 80Vdc, when a gap exists in the cable [7]. Conventional techniques such as the Fast Fourier Transform (FFT) are less effective in analyzing non-stationary signals [8]. Wavelet transformation has surfaced as a promising alternative, as it facilitates the concurrent analysis of signals in both time and frequency domains. In contrast to FFT, which only provides frequency data, wavelets can detect real-time signal changes, making them appropriate for identifying transient and nonperiodic arcing patterns. However, the selection of the optimal wavelet type remains a significant challenge. Recent studies, such as those by Shalby et al.[9]., demonstrate that the wavelet decomposition level significantly affects fault detection accuracy in distribution systems. In this study, we propose a systematic evaluation method that considers both detection accuracy and computational efficiency, making it highly suitable for real-time applications in systems with limited resources.

In its application, the Discrete Wavelet Transform (DWT) has frequently been utilized seriesto identify nonstationary signals or disturbances. In reference [10], a method for differential protection featuring adaptive threshold settings was introduced; however, the specifics of the wavelet filter employed were not thoroughly detailed. The properties of Series Arc Faults (SAFs) within direct current (DC) systems have been examined using the DWT, yet the optimal filter utilized remains unexplained [11]. In underground photovoltaic (PV) cable systems, fault localization has been executed through the use of wavelets and Artificial Neural Networks (ANN), albeit solely by decomposing the frequency sub-bands. In reference [12] the detection of failures in rotary machines was conducted using the DWT, but the best filter applied has not been established. In wind turbine systems equipped with lithium-ion batteries, the DWT, up to a decomposition level of 4, can identify fault

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points and reducing fluctuations [13]. In reference [14], wavelet transformation was employed on audio signals generated from an Internet of Things (IoT) system. Based on [15], Decomposition level 4 for DWT is the most effective value used to improve accuracy.

In the process of identifying disturbances, the Discrete Wavelet Transform (DWT) is frequently utilized in conjunction with machine learning (AI) and machine learning to enhance detection precision [16, 17]. A significant challenge associated with the application of DWT lies in the selection of the appropriate wavelet type and its corresponding decomposition level. This reliance on the level often necessitates a trial-and-error approach to ascertain optimal accuracy [18]. As noted in [19], the dependence on the mother wavelet and the decomposition level represents a limitation in reducing the likelihood of noise detection. Furthermore, it was elucidated in [20] that DWT exhibits considerable sensitivity to highly impulsive and transient disturbances, yet it remains heavily reliant on the chosen basis and decomposition level. Tan et al. [21] and Yin et al. [22] introduced sophisticated techniques that integrate feature extraction with machine learning to enhance both accuracy and robustness in the detection of series arc faults.

Numerous investigations have been undertaken regarding the integration of Discrete Wavelet Transform (DWT) with Artificial Intelligence (AI). Notable advancements have been achieved in reconceptualizing arc fault detection as an image recognition challenge through the application of a relative position matrix transformation [23]. The implementation of Hankel-Singular Value Decomposition (Hankel-SVD) for band-pass filtering significantly mitigates the interference caused by switching frequencies in photovoltaic systems [24]. Furthermore, the synergy between Linear Neural Networks (ANN) and Empirical Wavelet Transform (EWT) enhances fault classification within microgrids by facilitating improved feature extraction and analysis of residual currents [25]. The application of wavelet transform coupled with thresholding is also employed to evaluate Series Arc Faults (SAF), leading to enhanced accuracy in signal modeling. These advancements underscore the promise of wavelet filters and AI in augmenting the performance and dependability of systems that utilize diverse technologies [26,27]. According to the review presented in [28], persistent challenges exist in areas such as parameter selection, optimal wavelet basis determination, activation function choice in machine learning, implementation hurdles, and the depth of decomposition. Although wavelet-based techniques have been investigated across various fields such as fault detection, signal processing, and time-series analysis, our research makes a distinct contribution by methodically comparing several wavelet filters (Haar, Daubechies, Symlet, and Coiflet) within a controlled setting. Our methodology provides a more dependable, performance-oriented approach for selecting the most suitable wavelet filters for SAF detection, thereby ensuring both precision and computational efficiency. This contribution is in line with recent developments in the discipline, as illustrated by the work of Wu et al. [29], who combined wavelet transforms with deep learning frameworks to improve multi-scale feature extraction. Additionally, Shalby et al. [9] highlighted the significance of choosing the optimal decomposition level in wavelet analysis for fault detection in distribution networks. Zhao et al. [30] utilized dynamic wavelet techniques for predicting bearing life, demonstrating how dynamic wavelet decomposition can more effectively capture time-varying characteristics. Likewise, Gallegati [31] employed wavelet-based methods to analyze intricate climate data, further underscoring the adaptability and strength of wavelet transforms in examining non-stationary signals across various fields.

Given the numerous unresolved issues stemming from the challenges associated with wavelet selection, this research aims to evaluate wavelet filters to ascertain the most suitable type for specific applications involving difficult-to-detect SAFs. The methodology proposed in this research provides a systematic framework for selecting wavelet filters aimed at detecting Series Arc Faults (SAFs) in direct current (DC) systems. Traditional techniques often depend on a trial-and-error method for the selection of wavelet filters, which can prove to be inefficient. The study employs the Center of Gravity (CoG) method alongside radar chart visualization, utilizing wavelet parameter feature extraction. This approach markedly enhances the accuracy of SAF detection, especially in environments characterized by noise and transients. Prior research, including that of Wu et al. [29], has investigated the application of wavelet transforms in deep learning for multi-resolution analysis, which served as an inspiration for our evaluation technique. Moreover, when compared to previous methodologies, such as those proposed by Gallegati [31], which did not provide comprehensive assessments of wavelet filters, our approach guarantees a more systematic and reproducible evaluation of wavelet performance in the context of SAF detection. A comparative analysis is conducted on Haar, Daubechies, Symlet, and Coiflet wavelets to identify the most effective wavelet type. The choice of wavelets is predicated on the minimal computational resources required when employing the selected wavelet, thereby ensuring that the computational load for STM32F4 remains light and efficient. Reference [32] suggests that Meyer wavelets exhibit smooth values on signals, approaching infinity, which results in substantial computational demands. In contrast, bi-orthogonal wavelets are more commonly utilized for signal compression and reconstruction, rendering them inadequate for the detection of very short disturbances [33].

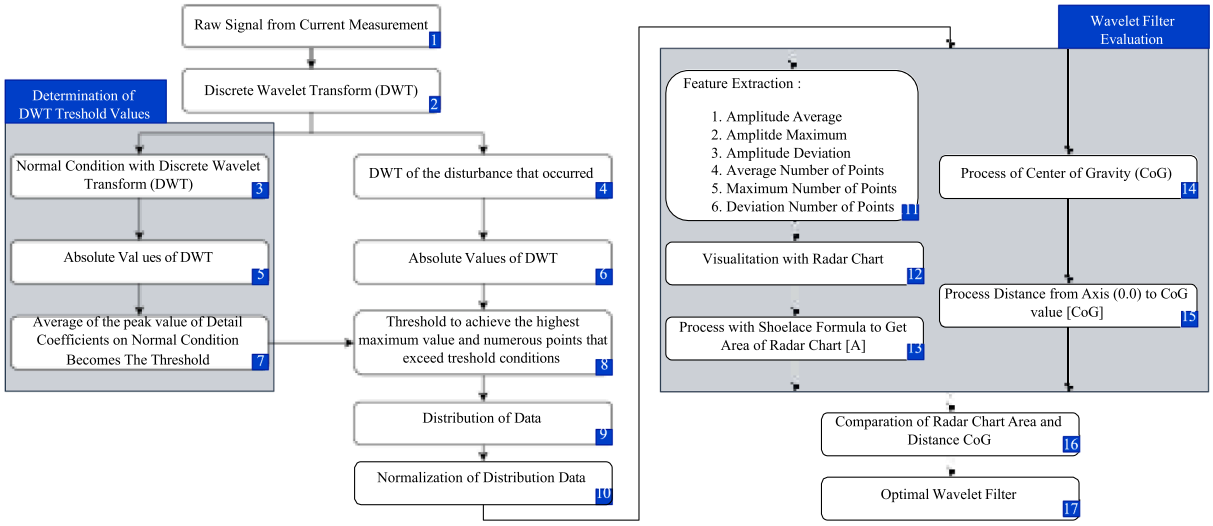
In the realm of recent advancements, our research presents a framework for performance-based wavelet filter selection that is applicable not only to SAF detection but also adaptable to various fault detection tasks. This method enhances the equilibrium between computational efficiency and detection performance, specifically tailored for real-time applications constrained by limited computational resources, such as systems based on microcontrollers. In contrast to current methodologies, which frequently lack a transparent and systematic evaluation process for filter selection, our approach addresses a significant gap in the literature by providing a more objective, repeatable, and practical solution for SAF detection.

SAFs are chosen due to their detection challenges, as they produce minimal current variations that often go unnoticed by protective devices [5]. Inverter noise further complicates detection, but variations in energy transfer linked to SAFs can be identified, potentially improved by inductive sensors [34]. This investigation underscores the significant challenges in fault detection, making SAFs a critical area of study [35, 36]. The current characteristics of series arc faults (SAF) are like noise and normal power fluctuations. Therefore, it is difficult to analyze based on the time domain. Thus, frequency domain-based method was chosen, namely the wavelet transforms. The contribution made by the authors is determining the threshold values that can be used to identify normal conditions, switching, and SAFs. The threshold value is limited to the wavelet methods used, namely Daubechies level 4, Symlet level 4, Coiflet level 4, and Haar level 4.

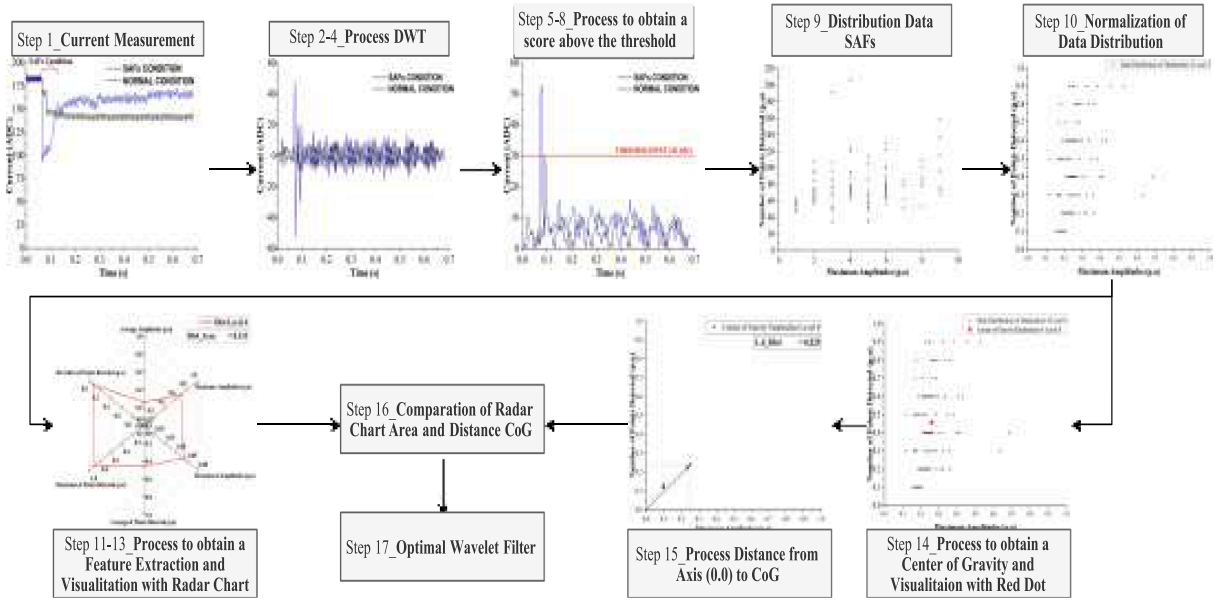
2. Proposed method of Sistematical evaluation

The Discrete Wavelet Transform (DWT) has surfaced as a promising approach due to its ability to analyze signals concurrently in both the time and frequency domains. Numerous studies have demonstrated the efficacy of wavelets in identifying stationary disturbances. The DWT proves to be more dependable in recognizing transient disturbances [37, 38], however, the selection of an inappropriate filter type may result in inaccuracies in disturbance detection. Fig. 1 illustrates the systematic methodology proposed in this research to identify the optimal filter for disturbance detection, utilizing the case study of SAFs within a PV system.

The implementation of the proposed method facilitates the identification of the most effective type of wavelet filter, particularly for use as a detection tool. As noted in [18], the selection of wavelet filters and types often relied on a trial-and-error approach, rendering the determination process challenging. The method proposed in this study offers a systematic approach to aid the selection of appropriate wavelet filters.



(a)



(b)

Fig. 1. Proposed method for optimal selection wavelet filter (a) flowchart and (b) visualization each stage.

2.1. Decomposition signal calculation

Wavelet signal decomposition acts like a movable window to analyze different frequency components of a signal. The equation for the Discrete Wavelet Transform (DWT) is presented in Eq. (1). In this context, $x(n)$ represents the input signal to be transformed, $\varphi(n)$ denotes the mother wavelet function, while a and b signify the dilation and translation parameters, respectively. The Mallat decomposition algorithm facilitates the DWT, dividing the signal into low-frequency (approximation) and high-frequency (detail) components iteratively until the desired detail level is reached. This procedure is reiterated until the desired decomposition level is reached. To ascertain the high-frequency value, we utilized Eq. (2), and for the low-frequency value, Eq. (3) was applied. For a decomposition level of 4, Equations (4) through (7) are applicable. In this context, $Y_{high}[k]$ represents the output of the high-pass filter (Mother Wavelet), $Y_{low}[k]$ indicates the output of the low-pass filter (detail coefficient), n corresponds to the number of points or wave data, g[n] is the low-pass filter, and h[n] is the high-pass filter.

The variation in the maximum value of k in (4)–(7) is caused by different wavelet filter lengths, which affect the DWT convolution range despite all using 4-level decomposition. Haar has the shortest filter with minimal signal extension, while Db4, Sym4, and Coif4 add +7, +3, and +11 coefficients, respectively, resulting in different output lengths after fourth-level downsampling and thus distinct maximum k values [39].

$$WT_{a,b}(n) = \frac{1}{\sqrt{a}} \sum_{0}^{N-1} x(n)\ \varphi\left(\frac{t-b}{a}\right) \tag{1}$$

$$Y_{high}[k] = \sum_{n=-\infty}^{n=\infty} x(n)\ h[2k-n] \tag{2}$$

$$Y_{low}[k] = \sum_{n=-\infty}^{n=\infty} x(n)\ g[2k-n] \tag{3}$$

$$\text{Factor for Haar level 4, } k = k = 0,\ 1,\ 2,\ \ldots,\ \left[\frac{N}{2^4}\right] - 1 \tag{4}$$

$$\text{Factor for Db 4 Level 4, } k = 0,\ 1,\ 2,\ \ldots,\ \left[\frac{N+7}{2^4}\right] - 1 \tag{5}$$

$$\text{Factor for Symlet 4 Level 4, } k = 0,\ 1,\ 2,\ \ldots,\ \left[\frac{N+3}{2^4}\right] - 1 \tag{6}$$

$$\text{Factor for Coiflet 4 level 4, } k = 0,\ 1,\ 2,\ \ldots,\ \left[\frac{N+11}{2^4}\right] - 1 \tag{7}$$

2.2. Detection treshold for DWT and feature extraction

In this research, the signal derived from the ADC measurement is referred to as $i[t]$, which is subsequently classified into normal current ($i_N[t]$) and current exhibiting arcing disturbances ($i_A[t]$). For each experimental scenario, the normal current data is denoted as $i_{Nc}[t]$. The detail coefficient $d_{Nc}[t]$ is acquired through a high-pass filtering technique utilizing wavelets, followed by the computation of its absolute value ($abs_d_{Nc}[t]$). The peak value of each detail coefficient is documented as $d_{max_N}[c]$.

$$i_{Nc}[t] = i_1, i_2 \cdots,\ i_t \text{ c}:\ 1,\ 2,\ \ldots, \text{c. c}\ :\ \text{case number} \tag{8}$$

$$d_{Nc}[t] =\ i_{Nc}[t]\ * h[n] \tag{9}$$

$$abs_d_{Nc}[t] =\ |d_{Nc}[t]| \tag{10}$$

$$d_{max_N}[c] =\ Max[abs_d_{Nc}[t]] \tag{11}$$

The arcing threshold is established based on the peak value of the DWT detail coefficient under standard conditions, incorporating a safety factor (k_s).

$$Threshold =\ ks \times Max[d_{max_N}[c]] \tag{12}$$

In This Investigation, a safety factor of 120% is applied. Results from under steady-state conditions, the maximum DWT coefficient ranges from 18 to 20 ADC, which after applying (k_s) remains below 30 ADC; therefore, the threshold is set at 30 ADC. This value represents the upper limit of normal operation, where any coefficient exceeding it is classified as arcing. Consequently, the threshold clearly distinguishes normal fluctuations from disturbance-induced transients (stages 5–8 in Fig. 1 (b)) and is defined based on Eqs. (8)–(12).

Following the acquisition of wavelet coefficients via 4-level decomposition, feature extraction is conducted to identify the characteristics of signal variations that surpass the detection threshold. This process entails the computation of statistical metrics such as average amplitude, maximum amplitude, amplitude deviation, and the count of detected points. The calculations incorporate standard deviation (σ), the total number of data points (n), individual data values (xi), and population mean (μ), as illustrated in Eqs. (13) and (14). The features extracted are subsequently utilized in later evaluation phases, which include normalization, center-of-gravity analysis, and radar chart comparison.

$$\sigma\ =\ \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2} \tag{13}$$

$$\mu = \frac{\sum x_i}{n} \tag{14}$$

2.3. Evaluation parameter & normalization

Normalization has been employed in research to evaluate the detected points against the highest ADC value obtained by the microcontroller. The peak normalization value is defined as one power unit (p.u.). This approach guaranteed a balanced comparison and facilitated the identification of the suitable wavelet type. The normalization formula is presented in Eq. (15) and Eq. (16).

$$Normalization\ of\ Number\ Detection\ Point(p.u) = \frac{Number\ Detection\ (n)}{Maximum\ number} \tag{15}$$

$$Normalization\ Current(p.u) = \frac{Maximum\ Current(ADC)}{Maximum\ Current\ Number} \tag{16}$$

2.4. Wavelet filter evaluation

2.4.1. Center of gravity (CoG)

The center of gravity in data distribution represents the average point that equilibrates all values, similar to the center of mass in physical entities. In the realm of statistics, this notion corresponds to the mean, which serves as the central point that minimizes the total squared distance. In the context of spatial data, it is computed as the average coordinate for the analysis of geographical distribution [40]. To ascertain the Centre of Gravity (CoG), one can utilize Eq. (17) for its calculation [41]. After obtaining the CoG value, the distance from the point (0,0) to the CoG can be determined using Eq. (17) and Eq. (18). Where x_i is the x-axis value, y_i is the y-axis value, m_i is weighting factor, n is number of data, and i is the i -th case or case number and d is the distance of CoG, x and y is CoG point in the x-y plane.

$$CoG = \left(\frac{\sum(x_i m_i)}{n},\ \frac{\sum(y_i m_i)}{n}\right) \tag{17}$$

$$d = \sqrt{x^2 + y^2} \tag{18}$$

2.4.2. Radar chart

Radar charts, or spider charts, are effective for comparing multiple quantitative variables simultaneously. They visualize data as a polygon with radial axes for each variable, making them useful for analyzing comparable data sets. In this research, radar charts illustrated the optimal wavelet transformation across six grids: average amplitude, maximum amplitude, amplitude deviation, average number of detected points, maximum number of detected points, and deviation from detected points. The results were analyzed using the Shoelace Formula, which calculates the area of a polygon by cross-multiplying coordinates, summing and subtracting the results, and dividing by two. This method accurately computes areas for various polygon shapes, including irregular ones, when coordinates are organized sequentially. The polygon area calculation method involves cross-multiplying x and y coordinates, summing and subtracting the results, and dividing by two. It works for various polygon types, including irregular shapes, as long as the vertices are in sequential order. The formula is expressed in Eq. (19), where A denotes the orthogonal area [42].

$$A = \frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1}) - \sum_{i=1}^{n}(y_i x_{i+1})\right| \tag{19}$$

3. Experimental setup for SAFs

An arc detection experiment was conducted following the UL1699 test standard to validate an integrated arc detection algorithm [43]. Fig. 2 illustrates a circuit diagram for the SAF testing. The setup included a circuit diagram for SAF testing, using a BK Precision MR50040 power supply to simulate photovoltaic conditions at 25 °C and 1000 W/m^2,

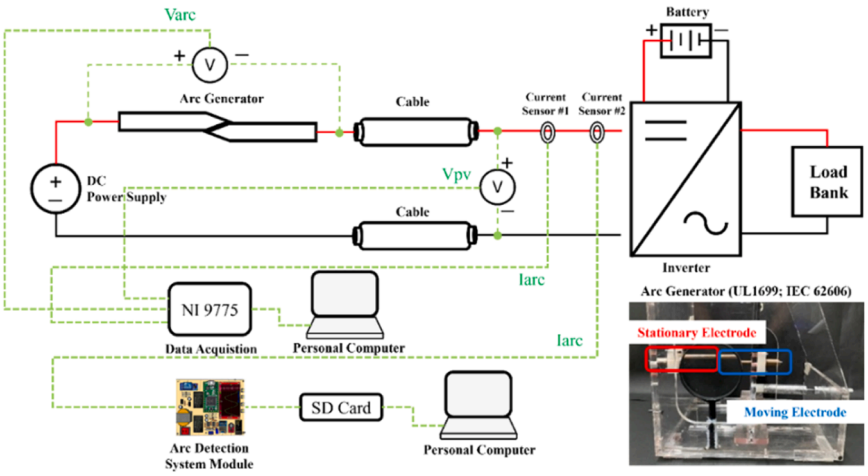


Fig. 2. UL 1699 Test Circuit for DC Series Arc Faults Detection.

with a maximum output of 5 kW, 500 V, and 40A Current measurements utilized various sensors: a Tektronix A622 DC current probe (100 kHz, 100A peak AC/DC), a Varc voltage sensor (2000 V, 20 MHz), and a Vpv voltage sensor for data acquisition via NI 9775 (±10 V, 20MS/s) for this measurement connected on NI 9775 (± 10 V, 20MS/s per channel). Additionally, a YHDC Hall effect sensor (100A AC/DC, 20 kHz) was connected to an STM32F4 microcontroller for data storage during SAF events. The experiment also included a Solana SOF12–100aH battery, a load bank with a maximum power of 5 kW at 50 Hz and Inverter manufactured by ICA Solar (SNV-GH3041 - 3kW/48 V, 450VDC/18A). Fig. 3 presents a photograph of the SAFs testing facility located at the high-voltage laboratory of the Institute of Technology Sepuluh Nopember (ITS) in Surabaya.

The study utilized four distinct loading configurations: 1300 W (100Vdc and 13A), 1500 W (110Vdc and 14A), 1800 W (130Vdc and 15A), and 2100 W (150Vdc and 15A). Additionally, two cable configurations were employed: ± 3 m and 50 m, to evaluate the effects of SAFs identified at different distances. In this setup, a total of 960 arcing tests were performed using an arc generator, which comprised a stationary electrode and a mobile electrode. The mobile electrode was set 2.5 mm apart and operated at a velocity of 5 mm per second to produce a series of arcs.

4. Result and analysis

4.1. The phenomenon of series arc faults

Arcing in resistive loads occurs when two conductors are separated by a gap, allowing air to conduct electricity. As depicted in Fig. 4, the arcing phenomenon initiates from the initial stationary electrode to the presence of a gap, resulting in arcing that halts once the electrode melts and creates a substantial gap. The illustration clearly indicates the prolonged duration during which SAFs in DC systems are challenging to eliminate, ultimately leading to the melting of copper. As a result, a significant gap is established, which leads to the cessation of arcing. Nevertheless, when arcing transpires, it can inflict damage on the surrounding area, potentially causing further harm to equipment connections such as junction boxes, cable duct fires, and similar incidents. The arcing phenomenon discharges energy between connections, generating a continuous arc that can attain temperatures of 3000 K. This phenomenon is primarily triggered by high voltage, which facilitates the creation of an electric arc through ionized air. The process encompasses ionization, plasma generation, and the release of energy in the form of light and heat. Arcing concludes when the electrodes melt, thereby increasing the air gap and limiting current flow, which illustrates its reliance on the physical conditions of the conductors. Fig. 6 shows the effect caused after arcing testing, where the electrode is oxidized to black or burnt, causing excessive heat throughout the electrode. Detection of Series Arc Faults (SAFs) may involve various measurements, such as current measurement (I_{arc}), voltage measurement (V_{arc}) at the arc fault generator, and PV voltage measurement (V_{pv}). As shown in Fig. 5, V_{arc} increase, while Iarc and Vpv decrease during SAF initiation. During SAF initiation, Vpv decreases, and when the arc reaches a stable condition, the PV voltage returns to normal because the lost voltage passes through the Varc measurement on the arc fault generator.

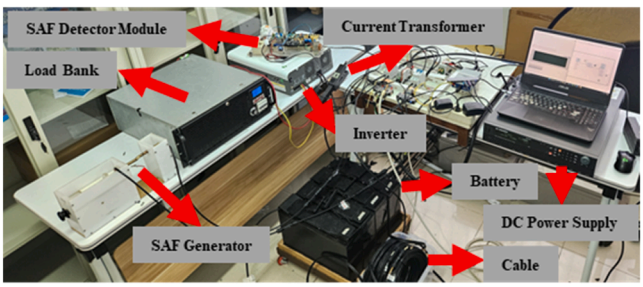


Fig. 3. Pictures of Test Facility for a Series Arc Faults Systems Detection.

4.2. Distribution data of SAFs

Arc fault signals (SAFs) in photovoltaic systems vary due to electrical arcing from issues like loose connections or damaged cables. The Discrete Wavelet Transform (DWT) decomposes these signals by frequency and time, isolating arc-related components from noise. DWT employs various wavelet bases (Haar, Daubechies, Symlet, Coiflet) to generate detail and approximation coefficients, with medium and high-scale coefficients indicating energy spikes from arc faults. This method enhances sensitivity to fault signals while ensuring computational efficiency, making it suitable for real-time monitoring and improving fault detection and system reliability. Fig. 7 illustrates the distribution of 120 arcing signal datasets that have been processed using various wavelet filters. Notably, the DB filter demonstrates a wider distribution compared to the others, emphasizing the critical role of wavelet selection in influencing the accuracy of SAF detection.

4.3. Evaluation wavelet filter using center of gravity

The center of gravity (COG) is a key principle in both physics and engineering, representing the location where an object's weight is uniformly distributed. Analyzing the COG is essential across multiple fields, including vehicle design, robotics, and biomechanics, to guarantee

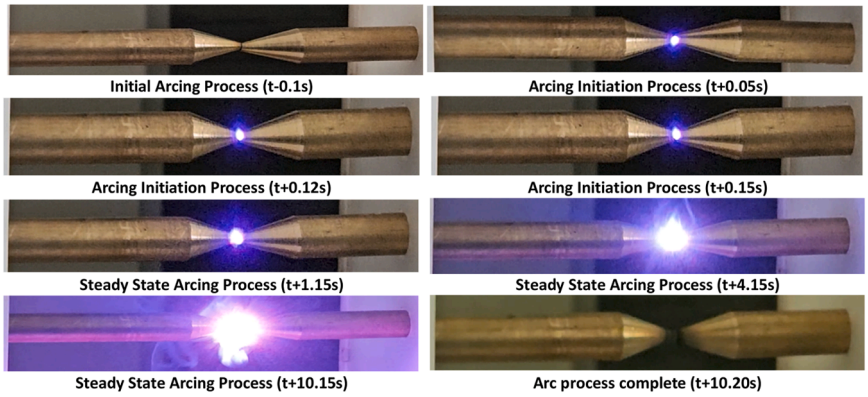


Fig. 4. Phenomenon of SAFs (start at time=t).

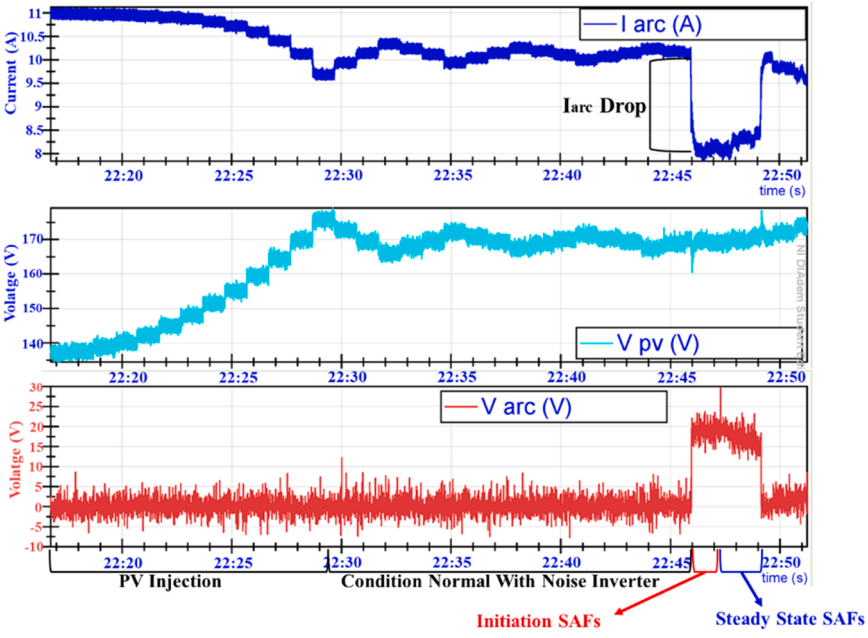


Fig. 5. Characteristic of series arc faults.

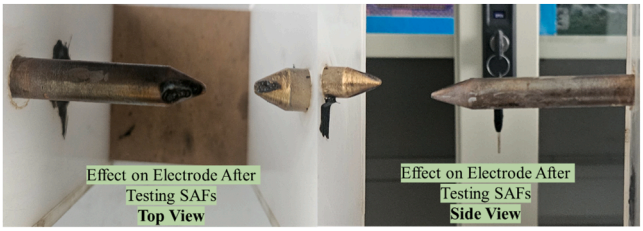


Fig. 6. Effects on Conductors after SAFs testing.

stability and efficiency. The calculations related to the COG are detailed in Eqs. (17) and (18). The COG value shown in Fig. 8 was determined using the black dot as the COG value. By applying this formula, one can ascertain the COG value; thereafter, the distance d from the point (0,0) to the COG will be computed to measure its magnitude. Fig. 9 shows the proposed COG values and this illustrates that the COG value is situated at the center of the data distribution spread. The COG results illustrated in Fig. 10, which suggest that Daubechies 4 Level 4 exhibits a distinctive pattern with a greater number of detection points than the other wavelet types, despite the amplitude values being generally lower. The image illustrates that the distance of the center of gravity (COG) ranges from the farthest to the nearest point relative to the coordinates (0,0). Thus, it is possible to determine the optimal wavelet filter.

The findings revealed a notable distinction between the two conditions. In the wireless scenario, the CoG values recorded were elevated across all wavelet types, suggesting that both the signal amplitude and the number of detected arcing points were superior to those observed in the wired scenario. In contrast, utilizing a 50-meter cable resulted in a decrease in the CoG value due to signal attenuation, which leads to a reduction in peak amplitude and a diminished count of successfully detected arcing points. This suggests that the length of the cable can influence the quality of the arcing signal captured by the detection system.

Regarding the performance of each wavelet, Daubechies (Db4) consistently yielded the highest CoG value, both with and without a cable, at 0.347 and 0.525, respectively. The Haar wavelet was positioned second, followed by the Symlet and Coiflet wavelets. This hierarchy indicates that Db4 exhibits superior sensitivity in capturing the characteristics of the arcing signal compared to the other wavelets. Furthermore, the disparity in CoG values between the cable and non-cable conditions was most pronounced for Db4, signifying that this wavelet can capture more signal information when not subjected to

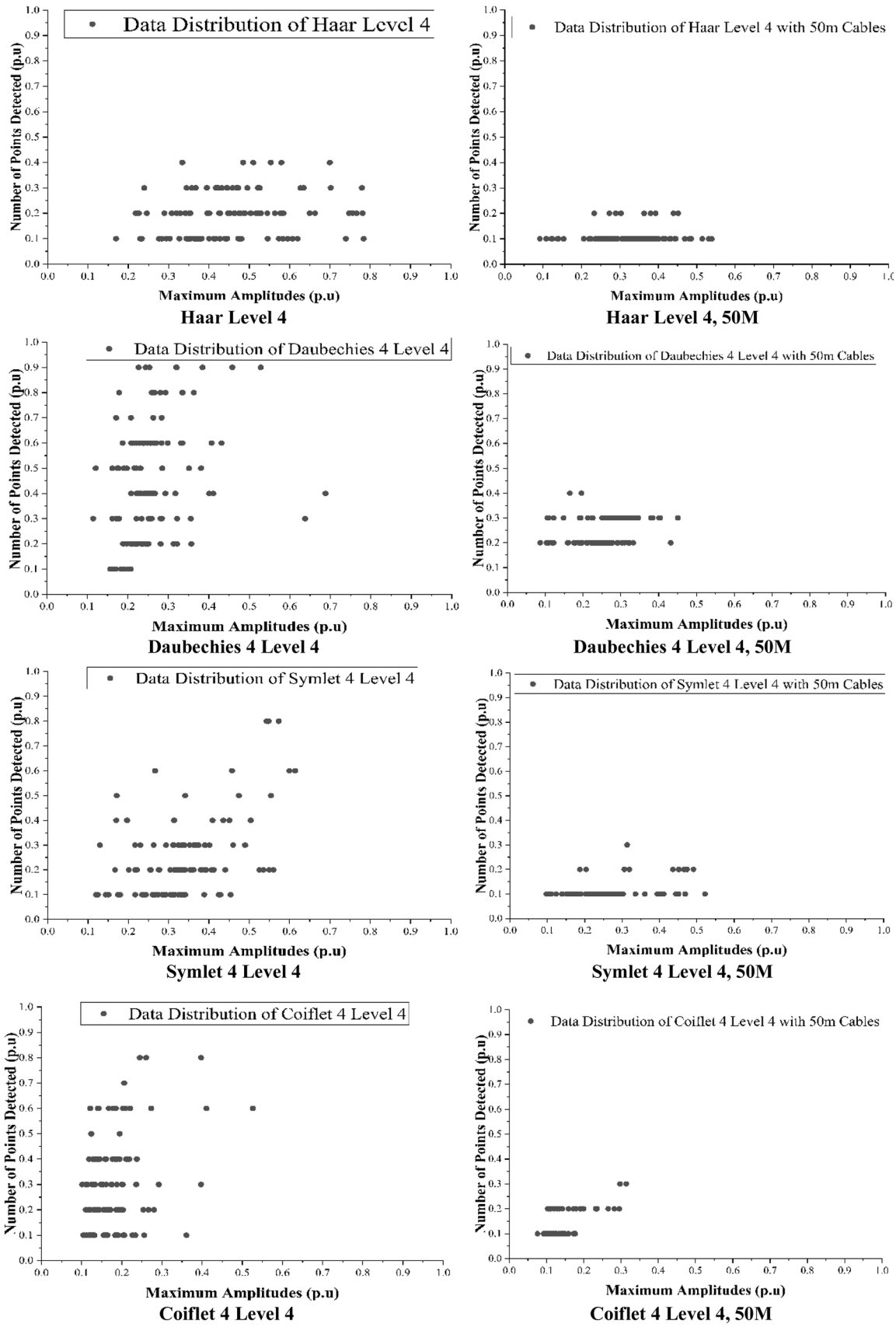


Fig. 7. Data distribution of SAFs.

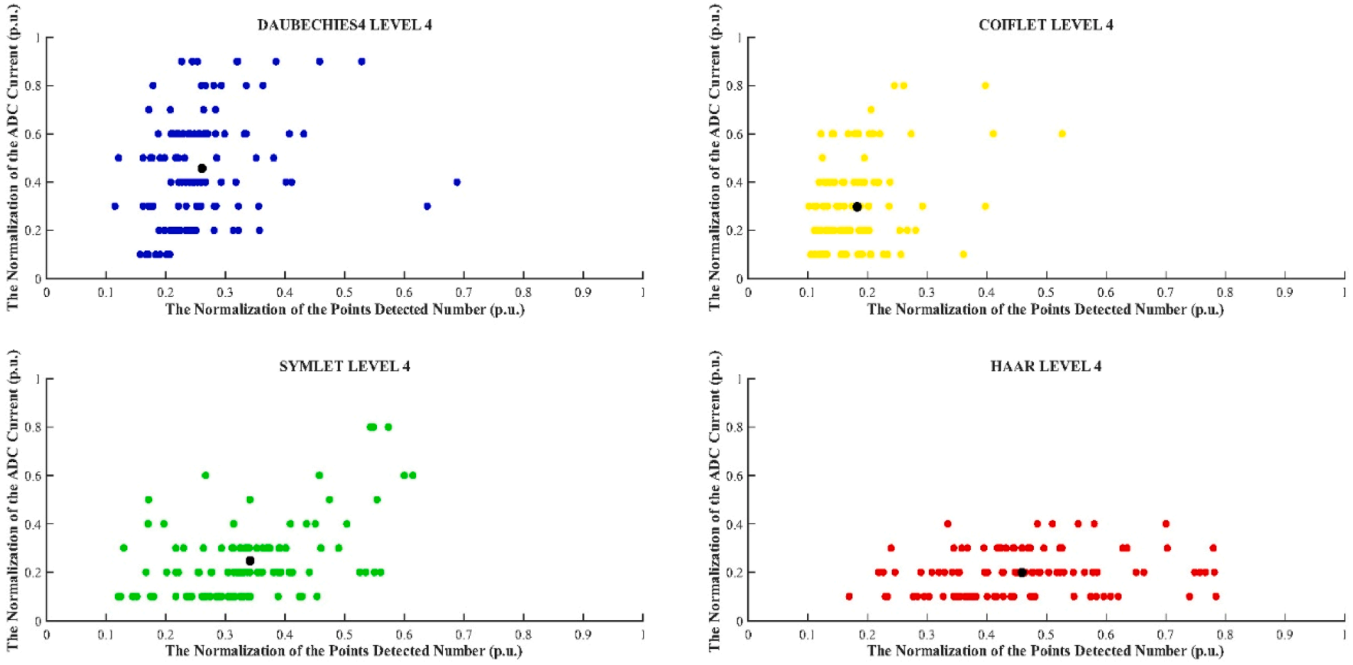


Fig. 8. Data distribution and center of gravity of SAFs on no cable.

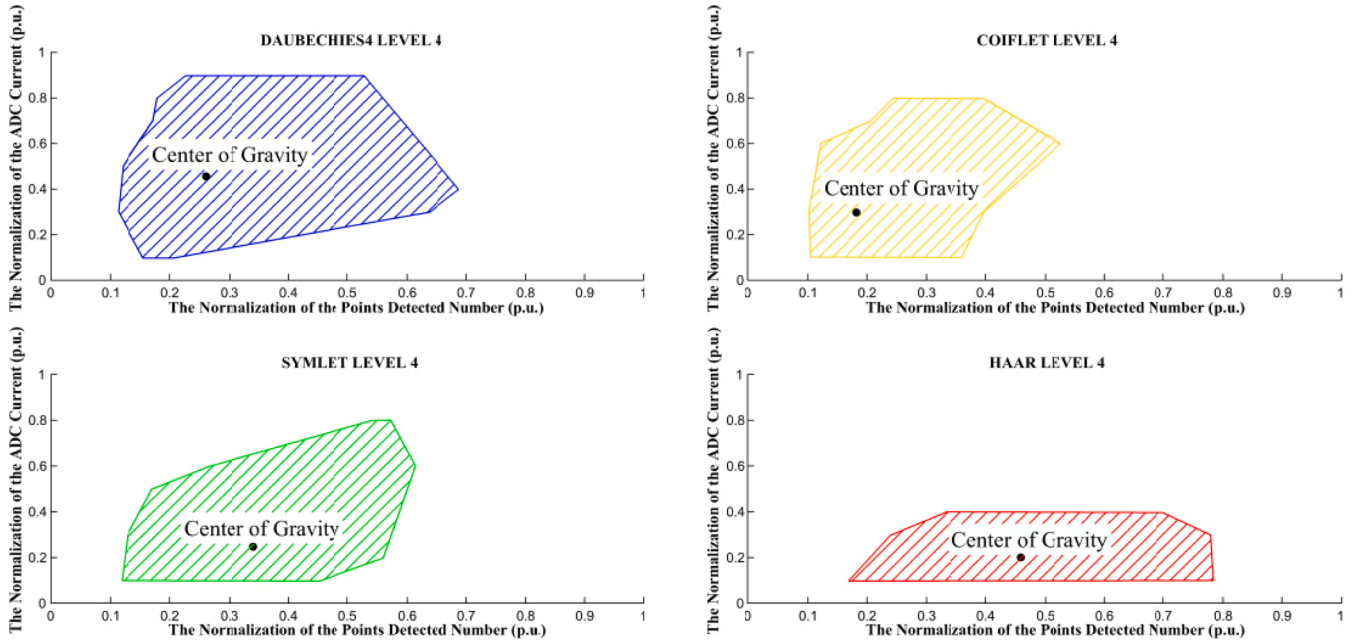


Fig. 9. Center of gravity of SAFs on no cable.

attenuation. This observation highlights the significance of selecting the appropriate wavelet type in early arcing detection systems, especially in systems employing cable installations of a specific length, such as photovoltaic (PV) systems. In this regard, the Daubechies 4 wavelet is advocated as the transformation method due to its ability to deliver optimal detection results despite performance degradation caused by cable attenuation. Consequently, factors related to signal attenuation and the choice of wavelet utilized must be critical considerations in the design of arc-monitoring systems based on signal analysis.

4.4. Evaluation wavelet filter using radar chart

A radar diagram was illustrated through the distribution of arcing in Fig. 11. This visual methodology allows for the intuitive identification of distribution patterns, consistency, and outliers, thus facilitating the interpretation of data complexity and offering an empirical foundation for discussing the study's results. Fig. 11 presents a radar diagram representing the arcing distribution. To ascertain the size of each wavelet area, the researcher employed the Shoelace formula as outlined in Eq. (19) to compute the area of an orthogonal shape.

According to the findings derived from the feature extraction radar graph based on the arcing data, a notable distinction was observed

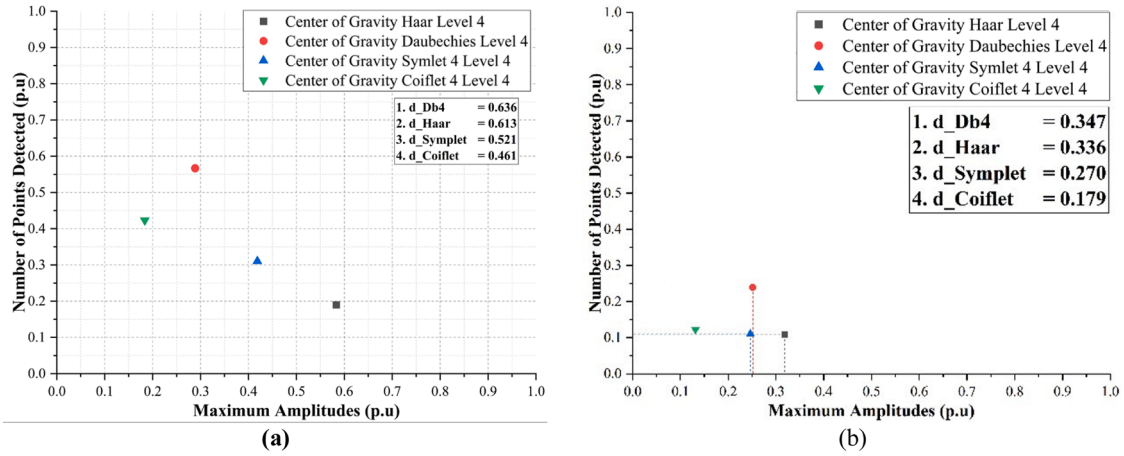


Fig. 10. Center of Gravity SAFs (a) no cable, and (b) with cable 50 m.

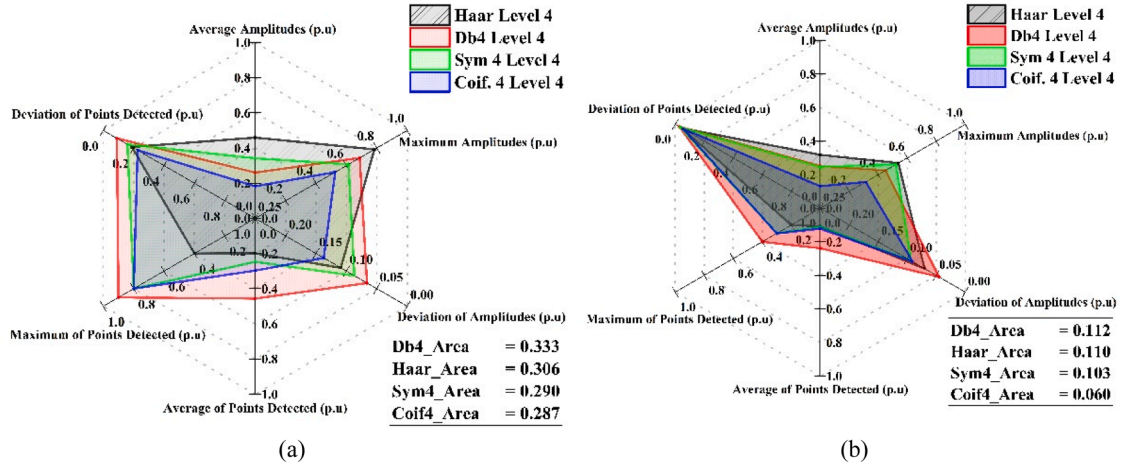


Fig. 11. Radar Chart SAFs (a) No cable, and (b) with cable 50 m.

between the scenarios with and without cables, particularly with a 50-meter cable extension. In the absence of cables, the wavelet Db4 level 4 method demonstrated superior performance, achieving an area of 0.333, followed by Haar (0.306), Sym4 (0.290), and Coif4 (0.287). This suggests that the arcing signals recorded under these circumstances possess more pronounced feature characteristics and are uniformly distributed across various parameters, including average amplitude, maximum amplitude, and the quantity and distribution of detected points.

Conversely, when the cable length was increased to 50 m, all wavelet methods experienced a marked reduction in performance, as evidenced by a decrease in the radar graph area. The most significant decline was noted in the Coif4 method, which only maintained an area of 0.060, indicating its heightened vulnerability to signal attenuation or distortion due to the extended cable length. In contrast, the Db4 and Haar methods displayed relatively stable performance, with areas of 0.112 and 0.110, respectively. This decline was particularly pronounced in the average number of detected points and the variability of the amplitude parameters, suggesting a diminishment in the distinctiveness or characteristic patterns of the arcing signal.

In summary, these findings underscore that cable length exerts a direct influence on the degradation of arcing signals, thereby impacting the efficacy of feature extraction through wavelet transformation. The highest area rankings in the absence of cables were Db4, Haar, Sym4, and Coif4. Under conditions involving a 50-meter cable, the area rankings, from highest to lowest, were Db4, Haar, Sym4, and Coif4.

5. Discussion

This paper presents a methodology for assessing the most suitable wavelet filter selection for scenarios. By employing this evaluation technique, the selection of wavelet filters can be implemented using an STM32F4 microcontroller. As indicated in [18], the process of selecting wavelet filters and their types frequently depends on a trial-and-error methodology, and the approach for choosing wavelet filters has not been extensively explored. Consequently, there is a deficiency of references regarding the selection of the optimal wavelet filter. SAFs in DC systems may manifest at low voltage levels, with a threshold voltage of 80 Vdc, which is sufficient to initiate arc faults.

Series arc faults (SAFs) present significant challenges for detection by circuit breakers due to their resemblance to normal operating conditions and the additional complexity introduced by inverter-generated noise. Nevertheless, variations in energy transfer associated with SAFs can be identified, potentially with the aid of inductive sensors [34]. As illustrated in Fig. 12, distinguishing arcing conditions from normal conditions is somewhat challenging. Even with the application of wavelet transformation, the differences become apparent, although the arcing curve exhibits a shape that is somewhat similar to fluctuations but with elevated values. The phenomenon of arcing within PV system manifests as a sudden and substantial increase in current relative to the standard operating conditions. In the graphs transformed by wavelet analysis, the arcing event is highlighted more distinctly, exhibiting sharp peaks that are absent in the typical current signal. The Haar waveform resembles

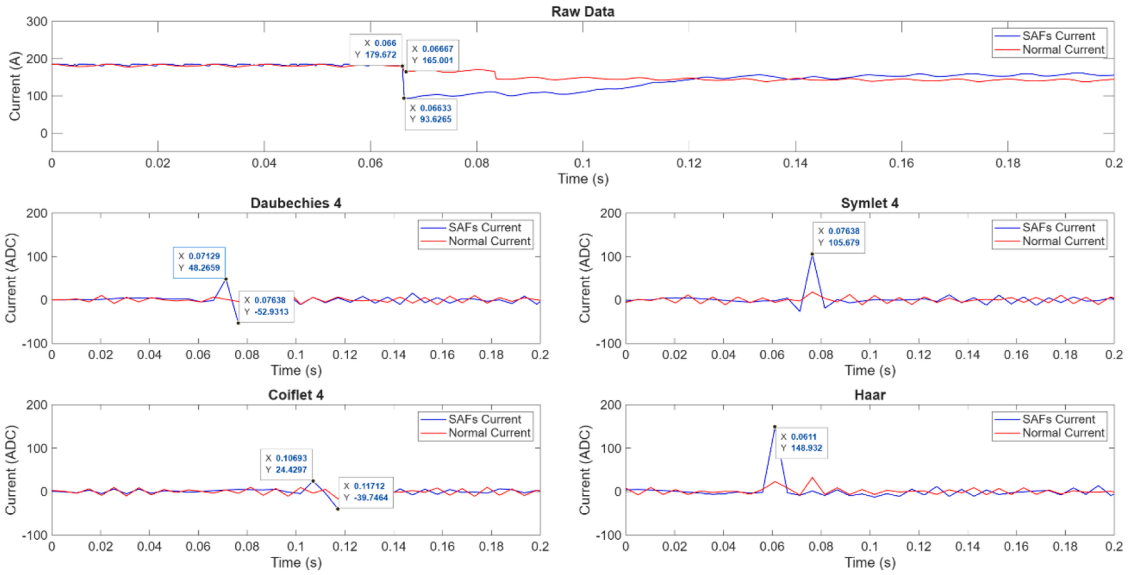


Fig. 12. Comparasion SAFs conditions and normal conditions.

that of the Daubechies, with the Haar transformation yielding a square-shaped result, while the Daubechies transformation produces a mountain-shaped result. If absolute values are considered, the peak values would differ significantly from those under normal conditions. The Symlet waveform is akin to the Coiflet waveform, with the transformation result manifesting as a three-mountain shape. The peak values in this wavelet type are smaller compared to those in the Haar/Daubechies transformation.

The limitations of this study are primarily associated with the restricted selection of wavelet filters, which is constrained by the hardware capabilities of the STM32F4 in detecting serial arc faults. Consequently, future research will explore alternative wavelet types for studies on serial arc faults. Currently, the selection of wavelet filters is confined to Haar, Daubechies, Coiflet, and Symlet, as these filters necessitate minimal computational resources, thereby facilitating the STM32F4′s computational processes. In terms of waveform analysis, Biorthogonal and Dmey wavelet types offer smoother characteristics; however, they demand more advanced computational capabilities.

The wavelet selection method outlined in this paper is notably effective owing to its systematic framework, which integrates objective performance metrics such as the Center of Gravity (CoG) and radar chart analysis. This approach facilitates a more organized assessment of wavelet filters, thereby removing the necessity for trial-and-error techniques typically employed in traditional methodologies. For instance, the Daubechies 4 (Db4) wavelet exhibited exceptional performance in identifying Series Arc Faults (SAFs), as demonstrated by its elevated CoG value and expanded radar chart area, which signify its enhanced capability to detect subtle signal variations compared to other filters such as Haar, Symlet, and Coiflet. This performance-driven, repeatable selection methodology guarantees that the selected filter is optimal not only for SAF detection but also for practical applications, where signal degradation due to cable length may influence detection precision. Consequently, the proposed method is deemed more dependable and robust in comparison to conventional techniques. In this research, we found that the Daubechies wavelet filter (Db4) consistently surpassed other wavelets, including Haar, Symlet, and Coiflet, in the detection of SAFs within a real-time context. The exceptional performance of Daubechies can be linked to its capability to capture transient characteristics in nonstationary signals, which is crucial for the precise identification of SAFs in power distribution systems.

6. Conclusion and future work

The Discrete Wavelet Transform (DWT) excels in recognizing nonstationary signal patterns, especially transient and nonperiodic arcing. The performance of wavelets is assessed through two key metrics: the Center of Gravity (CoG) and the area of the radar chart. An increased CoG distance signifies enhanced detection sensitivity, whereas a larger radar area suggests improved feature capture of arcing signals. The Db4 wavelet exhibits the most robust performance, attaining the highest CoG value of 0.406 and the largest radar area of 0.333, thereby establishing itself as the most efficient for identifying arc-related transients. By comparasing those two variables, the best performance of wavelet filter is able to be identified evethough the different is not significanly. Experimental findings suggest that the Daubechies4 (db4) level 4 method is the most efficient wavelet when compared to Haar, Symlet, and Coiflet, attaining approximately 4% higher scores in both CoG and Radar Chart assessments. The db4 filter consistently surpasses its counterparts in the analysis of nonstationary signals, and although the advantage over Haar is relatively modest, meticulous selection of wavelets is essential to avoid detection errors in sensitive applications. Taking into account the factors illustrated in Fig. 1 can aid in the optimal selection of the wavelet type and assess the effectiveness of wavelet filters in comparison to other wavelets. The constraints of this research are mainly linked to the limited variety of wavelet filters, which is restricted by the hardware capabilities of the STM32F4 in identifying serial arc faults. As a result, subsequent investigations will examine different types of wavelets for studies concerning serial arc faults.

Future investigations will concentrate on the real-time identification of disturbances within signals. By utilizing both the CoG and Radar Chart techniques, the most suitable wavelet can be identified. Furthermore, the detection mechanism alongside the wavelet evaluation method will aim to minimize detection errors. We also intend to implement this evaluation to ascertain the optimal wavelet for the real-time detection of SAFs using very light computing, namely microcontrollers. This enables highly effective arcing detection.

CRediT authorship contribution statement

Arief Budi Ksatria: Validation. **Dimas Anton Asfani:** Writing – original draft, Conceptualization. **I Made Yulistya Negara:** Writing – review & editing, Conceptualization. **Bagus Septianto:** Visualization. **Daniar Fahmi:** Data curation. **I Gusti Ngurah Satriyadi Hernanda:**

Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Dimas Anton Asfani, Doctoral reports was provided by Institut Teknologi Sepuluh Nopember. Dimas Anton Asfani, Doctoral reports a relationship with Institute Technology Sepuluh Nopember Department of Electrical Engineering that includes: employment and funding grants. Given his/her/their role as, had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] M. Erdiwansyah, R. Mamat, M.S.M. Sani, F. Khoerunnisa, A. Kadarohman, Target and demand for renewable energy across 10 ASEAN countries by 2040′, Electr. J. 32 (10) (2019) 106670 https://doi.org/10.1016/j.tej.2019.106670.
[2] IEA PVPS country updates 2024 - IEA-PVPS, Accessed: May 15, https://iea-pvps.org/annual-reports/annual-report-2024-country-updates/, 2025.
[3] J. Yan, Q. Li, S. Duan, A simplified current feature extraction and deployment method for DC Series arc fault detection, IEEE Trans. Ind. Electron. 71 (1) (2024) 625–634, https://doi.org/10.1109/TIE.2023.3247721.
[4] J.R. Riba, M. Moreno-Eguilaz, J.A. Ortega, Arc fault Protections for aeronautic applications: a review identifying the effects, detection methods, current progress, limitations, future challenges, and research needs, IEEE Trans. Instrum. Meas. 71 (2022) 1–14, https://doi.org/10.1109/TIM.2022.3141832.
[5] S.R. Madeti, S.N. Singh, A comprehensive study on different types of faults and detection techniques for solar photovoltaic system, Sol. Energy 158 (2017) 161–185, https://doi.org/10.1016/j.solener.2017.08.069.
[6] A. Amiri, H. Samet, T. Ghanbari, Recurrence plots based method for detecting series arc faults in photovoltaic systems, IEEE Trans. Ind. Electron. 69 (6) (2022) 6308–6315, https://doi.org/10.1109/TIE.2021.3095819.
[7] M.L. Eblen, T.A. Short, Low voltage ARC sustainability, IEEE Trans. Ind. Appl. 54 (2018) 2934–2946, https://doi.org/10.1109/TIA.2017.2776208.
[8] G.R. Venkatakrishnan, et al., Detection, location, and diagnosis of different faults in large solar PV system—a review', Int. J. Low-Carbon Technol. 18 (2023) 659–674, https://doi.org/10.1093/ijlct/ctad018.
[9] E.M. Shalby, A.Y. Abdelaziz, E.S. Ahmed, B. Abd-Elhamed Rashad, A comprehensive guide to selecting suitable wavelet decomposition level and functions in discrete wavelet transform for fault detection in distribution networks, Sci. Rep. 15 (1) (2025), https://doi.org/10.1038/s41598-024-82025-2.
[10] A.H.N. Tajani, A. Bamshad, N. Ghaffarzadeh, A novel differential protection scheme for AC microgrids based on discrete wavelet transform, Electr. Power Syst. Res. 220 (2023) 109292, https://doi.org/10.1016/j.epsr.2023.109292.
[11] S. Ouarhlent, N. Terki, M. Hamiane, H. Dahmani, Fault detection In robots based on discrete wavelet transformation and eigenvalue of energy, Diagnostyka 24 (4) (2023) 1–10, https://doi.org/10.29354/diag/174124.
[12] S. Sharma, S.K. Tiwari, S. Singh, Integrated approach based on flexible analytical wavelet transform and permutation entropy for fault detection in rotary machines, Measurement 169 (2021) 108389, https://doi.org/10.1016/j.measurement.2020.108389.
[13] A. Mannelli, F. Papi, G. Pechlivanoglou, G. Ferrara, A. Bianchini, Discrete wavelet transform for the real-time smoothing of wind turbine power using li-ion batteries, Energies 14 (8) (2021) 2184, https://doi.org/10.3390/en14082184.
[14] J.P.L. Escola, U.B. de Souza, R.C. Guido, I.N. da Silva, The Haar wavelet transform in IoT digital audio signal processing, Circuits. Syst. Signal. Process. 41 (7) (2022) 4174–4184, https://doi.org/10.1007/s00034-022-01979-8.
[15] S. Zhang, N. Qu, T. Zheng, C. Hu, Series arc fault detection based on wavelet compression reconstruction data enhancement and deep residual network, IEEE Trans. Instrum. Meas. 71 (2022) 1–9, https://doi.org/10.1109/TIM.2022.3158990.
[16] W. Dai, X. Zhou, Z. Sun, G. Zhai, Series AC arc fault detection method based on spectrogram and deep residual network, Microelectron. Reliab. 170 (2025) 115756, https://doi.org/10.1016/j.microrel.2025.115756.
[17] H. Hua, D. Fang, M. Guo, Series arc fault identification based on a novel Newton-optimized sparse kernel support vector machine, IEEE Trans. Instrum. Meas. 74 (2025) 1–13, https://doi.org/10.1109/TIM.2025.3579839.
[18] Q. Gong, K. Peng, Q. Gao, L. Feng, C. Xiao, Series arc fault identification method based on wavelet transform and feature values decomposition fusion DNN, Electr. Power Syst. Res. 221 (2023) 109391, https://doi.org/10.1016/j.epsr.2023.109391.
[19] S. Chen, X. Li, Y. Meng, Z. Xie, Wavelet-based protection strategy for series arc faults interfered by multicomponent noise signals in grid-connected photovoltaic systems, Sol. Energy 183 (2019) 327–336, https://doi.org/10.1016/j.solener.2019.03.008.
[20] Z. Wang, S. McConnell, R.S. Balog, J. Johnson, Arc fault signal detection - fourier transformation vs. wavelet decomposition techniques using synthesized data, in: 2014 IEEE 40th Photovoltaic Specialist Conference, PVSC 2014, 2014, https://doi.org/10.1109/PVSC.2014.6925625.
[21] M. Tan, Y. Gong, L. Wang, K. Li, J. Tong, Y. Su, LFArc-PFE: a series arc fault detection method based on low-frequency current data and perturbation feature extraction, IEEE Trans. Instrum. Meas. 74 (2025) 1–11, https://doi.org/10.1109/TIM.2025.3544714.
[22] Z. Yin, S. Peng, C. Xiao, L. Wang, S. Yang, A series arc fault detection method based on time-frequency markov permutation transition field for photovoltaic systems with power electronic devices, IEEE Trans. Power. Electron. 40 (8) (2025) 11380–11393, https://doi.org/10.1109/TPEL.2025.3555829.
[23] T. Zhang, R. Zhang, H. Wang, R. Tu, K. Yang, Series AC arc fault diagnosis based on data enhancement and adaptive asymmetric convolutional neural network, IEEE Sens. J. 21 (18) (2021) 20665–20673, https://doi.org/10.1109/JSEN.2021.3099638.
[24] X. Chen, W. Gao, C. Hong, Y. Tu, A novel series arc fault detection method for photovoltaic system based on multi-input neural network, Int J Electr Power Energy Syst. 140 (2022) 108018, https://doi.org/10.1016/j.ijepes.2022.108018.
[25] T. Biswal, S.K. Parida, S. Mishra, A combined approach of empirical wavelet transform and artificial neural network based high impedance fault classification in micro-grid system, in: 2022 IEEE Region 10 Symposium, TENSYMP 2022, 2022, https://doi.org/10.1109/TENSYMP54529.2022.9864362.
[26] C. Hu, N. Qu, S. Zhang, Series arc fault detection based on continuous wavelet transform and DRSN-CW with limited source data, Sci. Rep. 12 (1) (2022), https://doi.org/10.1038/s41598-022-17235-7.
[27] M. Farzanehdehkordi, S. Ghaffaripour, K. Tirdad, A.D. Cruz, A. Sadeghian, A wavelet feature-based neural network approach to estimate electrical arc characteristics, Electr. Power Syst. Res. 208 (2022) 107893, https://doi.org/10.1016/j.epsr.2022.107893.
[28] M.K. Alam, F. Khan, J. Johnson, J. Flicker, A comprehensive review of catastrophic faults in PV arrays: types, detection, and mitigation techniques, IEEE J. Photovolt. 5 (2015) 982–997, https://doi.org/10.1109/JPHOTOV.2015.2397599.
[29] J. Wu, J. Li, J. Yang, S. Mei, Wavelet-integrated deep neural networks: a systematic review of applications and synergistic architectures, Neurocomputing 657 (2025) 131648, https://doi.org/10.1016/j.neucom.2025.131648.
[30] J. Zhao, D. He, Z. Jin, X. Zhang, J. Zhou, A new method for bearing remaining useful life prediction based on dynamic wavelet and physical information constraints, Expert. Syst. Appl. 296 (2026) 129023, https://doi.org/10.1016/j.eswa.2025.129023.
[31] M. Gallegati, A systematic wavelet-based exploratory analysis of climatic variables, Clim. Change 148 (1–2) (2018) 325–338, https://doi.org/10.1007/s10584-018-2172-8.
[32] V. Valenzuela, H. Oliveira, Close Expressions for Meyer Wavelet and Scale Function, XXXIII Simpósio Brasileiro de Telecomunicações (SBrT2015) (2015), https://doi.org/10.14209/sbrt.2015.2.
[33] O. Ahmad, N.A. Sheikh, M. Ahmad, Nonuniform biorthogonal wavelets on positive half line via Walsh Fourier transform, J. Egypt Math. Soc. 29 (1) (2021), https://doi.org/10.1186/s42787-021-00128-5.
[34] J.B. Ahn, H. Bin Jo, H.J. Ryoo, Real-time DC series arc fault detection based on noise pattern analysis in photovoltaic system, IEEE Trans. Ind. Electron. 70 (10) (2023) 10680–10689, https://doi.org/10.1109/TIE.2022.3220883.
[35] S. Chen, X. Li, J. Xiong, Series arc fault identification for photovoltaic system based on time-domain and time-frequency-domain analysis, IEEE J. Photovolt. 7 (4) (2017) 1105–1114, https://doi.org/10.1109/JPHOTOV.2017.2694421.
[36] E. Calderon-Mendoza, P. Schweitzer, S. Weber, A double ended AC series arc fault location algorithm for a low-voltage indoor power line using impedance parameters and a neural network, Electr. Power Syst. Res. 165 (2018) 84–93, https://doi.org/10.1016/j.epsr.2018.08.008.
[37] A. Mellit, G.M. Tina, and S.A. Kalogirou, 'Fault detection and diagnosis methods for photovoltaic systems: a review', 2018. doi: 10.1016/j.rser.2018.03.062.
[38] H. Zhao, J. Liu, J. Lou, Series arc fault detection based on current fluctuation and zero-current features, Electr. Power Syst. Res. 202 (2022) 107626, https://doi.org/10.1016/j.epsr.2021.107626.

[39] Wavelet analysis and its applications, Wavelet Anal. Appl. 1 (100) (1992) ii, https://doi.org/10.1016/B978-0-12-174584-4.50001-3. Jan.
[40] R.C. Hibbeller, Engineering Mechanics Statics 12th Edition, Prentice Hall, 2010.
[41] J. Gechlik, H. Sedrakyan, Gauss's area formula for irregular shapes, Ohio J. Sch. Math. 97 (1) (2024), https://doi.org/10.18061/ojsm.4172.
[42] B. Braden, The surveyor's area formula, Coll. Math. J. 17 (4) (1986) 326–337.
[43] D.A. Dini, P.W. Brazis, K.H. Yen, Development of arc-fault circuit-interrupter requirements for photovoltaic systems, in: Conference Record of the IEEE Photovoltaic Specialists Conference, 2011, https://doi.org/10.1109/PVSC.2011.6186301.